

POLARIS DC-1 — ELV, BMS, FIRE DETECTION AND SUPPRESSION

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Project: Polaris DC-1 — Aviapolis Data Campus, Phase 1, Vantaa, Finland
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Synthetic demonstration data. All parties, ratings and figures are fictional.

1. Scope

- 1.1 This specification covers the design, supply, installation, testing and commissioning of the extra-low-voltage systems at Polaris DC-1: the Building Management System (BMS), Data Centre Infrastructure Management (DCIM) reporting layer, fire detection, fire suppression, and the security/access-control and CCTV systems.
- 1.2 The Subcontractor shall carry design responsibility for the means of compliance with PDC-ER-001 §5 and §6, using PDC-BOD-001 §8 as the design intent to be carried forward unchanged unless agreed under the S1–S3 review process.
- 1.3 Excluded: mechanical and electrical plant itself (refer PDC-SPEC-05 and PDC-SPEC-06 for equipment schedules — this specification covers the field devices, controls network, detection, suppression and security systems that monitor and protect that plant); Tenant-supplied cage/cabinet-level access control hardware (PDC-TA-001 §11); the Tenant's credential database (Tenant-supplied, integrated at handover).
- 1.4 White-space suppression method (inert gas or water mist) is a **fire-engineering submission item** subject to authority approval and is not fixed by this specification; the Subcontractor shall carry forward whichever method is approved at S2 detailed design without variation to the performance requirements below.

2. Standards

Reference	Application
National Building Code of Finland	Fire safety chapters, means of escape
Land Use and Building Act (132/1999)	Statutory design basis
Rescue Act (379/2011)	Fire and rescue service requirements, authority approval of the fire-engineering submission
SFS 6000	Low-voltage installations — ELV supplies, control panels
EN 50600 series	Data centre facility and management/operational information design reference
EN 54 series	Fire detection and fire alarm systems, including aspirating smoke detection (EN 54-20)
EN 15004 series	Fixed firefighting systems — gaseous extinguishing systems (if inert gas selected)
NFPA 750 (reference)	Water mist systems, referenced where inert gas is not selected, subject to authority acceptance
ISO 22320	Emergency management reference for incident response coordination
PDC-ER-001 §5 and §6	Design intent for fire-engineering submission

BACnet/IP (ISO 10484-3)	Open protocol for BMS field and supervisory network
IEC 62443	Industrial network/control system cybersecurity, applied to the BMS/DCIM network
EN 50131 series	Intrusion and access control systems
EN 62676 series	CCTV systems for use in security applications

Where standards conflict, the more onerous requirement governs, per PDC-BOD-001 §1.2.

3. Equipment schedules

3.1 BMS and DCIM

Item	Requirement
Protocol	Open protocol, BACnet/IP throughout, no single-vendor lock-in
Coverage	All mechanical and electrical plant across all four halls, substation, generator compound and cooling yard
DCIM	Per-hall and per-feed power, cooling and environmental reporting, sitting above the BMS
Historical trending	Retained not less than 24 months
Metering integration	Reads per-hall/per-feed energy metering and per-rack Class 0.5S metering from PDC-SPEC-06
Network	Dedicated, segregated control network, IEC 62443 zone/conduit model applied

3.2 Fire detection

Item	Requirement
White space	Aspirating smoke detection (EN 54-20), minimum two detection zones per hall
Other areas	Addressable point detection to EN 54, National Building Code coverage density
Battery rooms	Off-gas / thermal-runaway detection ahead of the aspirating smoke system, per PDC-SPEC-06 §3.3
Panel	Addressable fire alarm control panel per Section, networked to a site-wide repeater at the gatehouse/NOC

3.3 Fire suppression

Item	Requirement
White space	Inert gas or water mist, subject to fire-engineering approval by the authority — final selection fixed at S2
Battery rooms	Suppression method coordinated with the lithium-ion fire strategy, no water-based suppression directly onto energised battery racks
Other areas	Sprinkler to National Building Code, dry riser where required by the authority
Compartmentation	Each hall a separate fire compartment, EI 120 minimum, continuous foundation to roof deck

3.4 Security and access control

Item	Requirement
Access control layers	Five layers: perimeter, building entry (mantrap), facility floor, hall entry, cage/cabinet row (Tenant-managed)
Perimeter	Fence line and intrusion detection around the full plot, ANPR vehicle gate at the single site access point
Credentialing	Card/biometric, integrated with the Tenant's credential database at each Section's handover
CCTV coverage	Perimeter, gatehouse, spine circulation, each hall entry
CCTV retention	Not less than 90 days
Escort logging	System capable of recording escort status for uncredentialed personnel per hall/layer

4. Installation and workmanship

- 4.1 Aspirating smoke detection pipe networks shall be installed within the coordinated overhead service zone, with sampling points positioned per the manufacturer's design software output and submitted for Engineer review at S2.
- 4.2 All fire alarm and suppression cabling shall be fire-rated and segregated from power cabling in accordance with the National Building Code and the manufacturer's installation instructions.
- 4.3 No penetration of an EI 120 fire compartment wall shall be made without a fire-stopping detail achieving the same rating, witnessed and photographically recorded before the wall is closed up.
- 4.4 BMS and DCIM network cabling shall be installed on a physically segregated containment run from the security/access-control and CCTV network, with both networks isolated from the Tenant's own IT network per IEC 62443 zone/conduit principles.
- 4.5 Access control readers and CCTV cameras shall be installed and commissioned hall-by-hall so that each hall's security systems are complete and independently testable at that hall's Section handover.
- 4.6 All field devices (detectors, dampers, valves, readers) shall be labelled with a unique tag matching the BMS point list and the O&M schedule.
- 4.7 Suppression system discharge testing (or equivalent non-destructive verification accepted by the authority) shall be witnessed and recorded before Section handover.

5. Control and sequences of operation

5.1 Cause-and-effect summary (extract)

Cause	Effect
Aspirating smoke detection, Alert level, single zone	BMS alarm to NOC; DCIM log; no plant shutdown
Aspirating smoke detection, Fire 1, single zone	Fire alarm sounders activate in the affected hall; BMS alarm to NOC; suppression system arms
Aspirating smoke detection, Fire 2, confirmed	Suppression release sequence initiated (subject to pre-discharge warning period per the approved fire-engineering submission); AHU/CRAH shutdown and fire/smoke damper closure in the affected hall only; other halls unaffected
Battery room off-gas detection	Isolate affected UPS module; suppression method per lithium-ion fire strategy; NOC and FM notification; adjacent modules remain in service
Manual call point activated	Fire alarm sounders activate in the affected zone; BMS alarm to NOC
Loss of fire alarm panel communication	BMS critical alarm; local panel continues to operate in standalone mode
Perimeter intrusion detection triggered	CCTV auto-recall to the affected zone; security alarm to NOC; access log entry created
Access denied, five failed attempts	Local alarm; security notification; access log entry created
Mantrap door held open beyond timeout	Local alarm; security notification

5.1.1 The full cause-and-effect matrix shall be developed at S2 detailed design, agreed with the Employer and the authority before commissioning, and witnessed as part of L3/L4 commissioning per PDC-ER-001 §6, with no unassigned critical alarms outstanding at handover.

5.2 Alarm hierarchy

Priority	Response time (NOC acknowledgement)	Examples
Critical	Immediate	Fire 2 confirmed, suppression discharge, loss of N+1 on any monitored system
High	≤ 5 minutes	Fire 1 single zone, single component fault with redundancy intact
Warning	≤ 30 minutes	Sensor drift, filter differential pressure high
Informational	Logged only	Scheduled maintenance mode entry/exit

5.3 Setpoints (extract, cross-referenced to PDC-SPEC-05)

Parameter	Setpoint	Source
Aspirating smoke detection sensitivity	Per manufacturer's design software, two-stage (Alert/Fire)	This specification
White space temperature/humidity alarm	24 °C ± 1.5 K, 20–70% RH	PDC-SPEC-05 §5.2
CCTV recording resolution and frame rate	Per Engineer-approved schedule, minimum to support 90-day evidentiary retention	This specification

6. Testing and commissioning

Level	Activity	ELV/fire/security scope
L1	Factory acceptance testing	Fire alarm panels, suppression control panels — witnessed functional test
L2	Site delivery and static inspection	Visual inspection, device nameplate verification against schedule
L3	Individual component start-up and functional test	Each detector zone, damper, access reader, CCTV camera tested individually
L4	Integrated systems testing, including black-building and failure-mode tests	Full cause-and-effect matrix witnessed per hall; integrated systems test corresponding to PDC-TA-001 §4
L5	Integrated load-bank testing to full Section load for 24 hours continuous	BMS/DCIM monitoring of the full load-bank test, alarm-free operation demonstrated throughout

6.1 No unassigned critical alarm shall remain outstanding at Section handover, per PDC-ER-001 §6.

6.2 Suppression discharge testing (or authority-accepted equivalent) shall be completed and witnessed before the Section's occupancy inspection (käyttöönottotarkastus).

7. Submittals

7.1 The Subcontractor shall submit, at the design stages defined in PDC-ER-001 §10:

- fire-engineering submission for white-space suppression method selection, for authority approval at S2;
- full cause-and-effect matrix, for Employer agreement before commissioning;
- BMS/DCIM point list and network architecture, including IEC 62443 zone/conduit diagram;
- CCTV camera schedule and coverage plan, confirming no coverage gap at perimeter, entry or hall-entry points;
- access control credentialing integration plan, coordinated with the Tenant's credential database format.

8. Spare, training and warranty

8.1 The Subcontractor shall supply a spares schedule covering not less than two years of routine maintenance consumables (detector heads, aspirating filters, suppression cylinders where applicable) and one critical spare of each fire alarm panel type and access control reader type.

8.2 The Subcontractor shall provide not less than 5 days of operator training per Section, covering BMS/DCIM operation, fire alarm panel operation, cause-and-effect response and security system administration, with training records forming part of the S4 handover documentation.

8.3 Equipment warranties shall run for not less than 24 months from the relevant Section's Service Commencement date, or 30 months from delivery, whichever is later, and shall be assignable to the Employer's facilities management provider.

8.4 The Subcontractor shall maintain critical-plant response cover (fire alarm, suppression, access control) for the defects period, with a maximum 4-hour attendance time for a Critical alarm per Section 5.2.

End of PDC-SPEC-07 Rev A.